Week 6 Date:15/10/2026

Propositional Logic

```python
# Variables: A, B, C
# alpha = A V B
# KB = (A V C) ^ (B V ~C)


def to_str(val):
  return "true" if val else "false"


# Headers aligned exactly with the table columns
print(f"{'A':<7} {'B':<7} {'C':<7} || {'A V C':<7} | {'B V ~C':<7} | {'KB':<7} | {'alpha'}")
print("=" * 65)

# Setting values as [False, True] matches the false-first ordering in the image
values = [False, True]

for A in values:
  for B in values:
    for C in values:
      # Intermediate column calculations
      a_or_c = A or C
      b_or_not_c = B or (not C)

      # Knowledge Base and Alpha
      kb = a_or_c and b_or_not_c
      alpha = A or B

      # Printed with consistent column spacing and lowercase values
      print(
          f"{to_str(A):<7} {to_str(B):<7} {to_str(C):<7} || "
          f"{to_str(a_or_c):<7} | {to_str(b_or_not_c):<7} | "
          f"{to_str(kb):<7} | {to_str(alpha)}"
      )
```

Output:

```
A       B       C       || A V C   | B V ~C  | KB      | alpha
=================================================================
false   false   false   || false   | true    | false   | false
false   false   true    || true    | false   | false   | false
false   true    false   || false   | true    | false   | true
false   true    true    || true    | true    | true    | true
true    false   false   || true    | true    | true    | true
true    false   true    || true    | false   | false   | true
true    true    false   || true    | true    | true    | true
true    true    true    || true    | true    | true    | true
```